Early blood pressure management in acute ischaemic stroke before and after reperfusion treatment

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Lancet Neurol 2026; 25: 206–12

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Summary

Background **Patients with acute ischaemic stroke often present with elevated blood pressure. Evidence has emerged that challenges the longstanding recommendation across clinical guidelines that blood pressure should be controlled to a systolic pressure of less than 185 mm Hg before initiating reperfusion therapy in these patients.**

Recent developments **Major clinical trials (INTERACT4, MR-ASAP, and RIGHT-2) and an observational study (TRUTH) indicate that intensively lowering blood pressure before initiating reperfusion therapy for acute ischaemic stroke is associated with increased risk of death and disability. Similar evidence has reshaped our approach to blood pressure management in patients who receive successful endovascular thrombectomy for acute ischaemic stroke from large-vessel occlusion. The underlying mechanisms of cerebral ischaemia are likely to involve alterations in cerebral blood flow through collateral vessels and the microcirculation.**

Where next? **Current evidence underscores the harms of more intensive lowering of blood pressure to a systolic target of less than 140 mm Hg both before and after reperfusion therapy for acute ischaemic stroke. Despite the challenges in obtaining the necessary randomised evidence, such efforts are now required to establish whether elevated blood pressure should be controlled at all with this disease or left to decrease naturally.**

Introduction

Acute ischaemic stroke is the most common type of stroke and the second leading cause of death worldwide.[1,2] Reperfusion therapy with intravenous thrombolysis and endovascular thrombectomy is well established as an effective treatment.[3,4] However, many patients do not have good functional recovery from their illness due to the limitations of reperfusion therapy and the complex nature of the evolving ischaemic penumbra.[4,5] The management of co-occurring elevated blood pressure in patients with acute ischaemic stroke has long been suggested, albeit cautiously, as a readily modifiable approach to reduce the likelihood of secondary ischaemic injury, haemorrhagic transformation, and cerebral oedema after reperfusion therapy.[6–11] However, because reducing blood pressure too precipitously can compromise cerebral perfusion when the brain is dysregulated by cerebral ischaemia,[12–14] guidelines have persisted in recommending a systolic blood pressure of less than 185 mm Hg before and 180 mm Hg after the initiation of reperfusion therapy.[15–17] Despite being chosen somewhat arbitrarily, these blood pressure threshold criteria have long served as reasonable recommendations retained from the protocol for the National Institutes of Health studies that established intravenous alteplase as an effective treatment of acute ischaemic stroke.[18] These criteria have subsequently been widely applied in the licensing approval for intravenous alteplase, in further trials of reperfusion therapy, and then expanded into recommendations for the use of mechanical thrombectomy.[19–23]

However, during 2022–25, several randomised controlled trials[24–30] and a multicentre prospective observational study[31] raised doubts as to whether these criteria for blood pressure control should continue to be applied. In this Rapid Review, we provide an overview of the latest evidence on the management of blood pressure before reperfusion therapy and specifically after successful endovascular thrombectomy in acute ischaemic stroke. We conclude by summarising key outstanding issues for future research that must be addressed to advance the field.

Recent research developments

The management of blood pressure before reperfusion therapy

The effectiveness of lowering blood pressure was evaluated in the INTERACT4 trial[24] (bolus of intravenous urapidil) and the MR-ASAP trial[25] (transdermal application of a 5 mg glyceryl trinitrate patch for a 24 h duration) in patients with suspected acute stroke assessed in ambulance settings. Both trials showed a strong signal of harm in patients with acute ischaemic stroke within the eligible time window for receipt of reperfusion therapy. MR-ASAP,[25] which was conducted at six ambulance service centres that served 18 hospitals in the Netherlands, was prematurely terminated in June, 2021—after just 380 patients were recruited—on the recommendation of the data and safety monitoring board, who were concerned over safety in the subgroup of patients with acute intracerebral haemorrhage. The data reflected a similar signal of harm in the earlier and larger RIGHT-2 trial with prehospital glyceryl trinitrate transdermal application,[32] which showed an overall neutral result for functional outcome on the modified Rankin scale (mRS) for patients assessed at 90 days (adjusted common odds ratio [OR] 0·97 [95% CI 0·65–1·47]). However, in MR-ASAP, the point estimate

was associated with less favourable functional outcomes in the glyceryl trinitrate group compared with the no-treatment control group (0·67 [0·39–1·13]) in 201 (62%) of 325 patients with acute ischaemic stroke (with or without reperfusion therapy) who provided deferred consent.[25] However, the signal for harm was stronger in those patients who went on to receive intravenous thrombolysis (0·51 [0·26–0·99]). Although the simple fixed-dose approach to treatment in MR-ASAP[25] and RIGHT-2[32] differed from the typical titratable, treat-to-target use of intravenous antihypertensive agents in hospital-based clinical practice, the consistent finding of potential harm from even a small reduction in systolic blood pressure raised concerns in the design and conduct of the INTERACT4 trial.[24] The key characteristics and outcomes of these three prehospital trials are shown in the appendix (p 2).

INTERACT4 enrolled 2404 patients in ambulances who had a high likelihood of acute stroke and a systolic blood pressure of 150 mm Hg or more within 2 h of the onset of symptoms.[24] Doctors employed by ambulance services in participating regions of China randomly assigned patients to receive either intravenous urapidil to reduce systolic blood pressure to a target of 130–140 mm Hg within 30 min, or to avoid treating blood pressure unless it was extremely elevated (≥220 mm Hg) until the patient arrived at hospital. The mean between-group difference in systolic blood pressure was –13·6 mm Hg (95% CI –16·0 to –11·1 mm Hg) in the ambulance. Patients with acute ischaemic stroke in the group treated with urapidil had higher odds of poor functional outcome according to the full range of scores on the mRS (common OR 1·30 [95% CI 1·06–1·60]) and a greater 90-day mortality (23% *vs* 17%; OR 1·48 [95% CI 1·10–2·00]) compared with patients in the group with avoided treatment. Of the patients with cerebral ischaemia, 274 (44·6%) of 614 in the intervention group and 256 (41·8%) of 612 in the usual-care group received intravenous thrombolysis. Also in patients with cerebral ischaemia, 75 (12·2%) of 614 patients in the intervention group and 61 (10·0%) of 612 patients in the usual-care group underwent endovascular thrombectomy.[24] Despite variability in the approach to blood pressure lowering used in these three ambulance trials, the results are consistent in indicating adverse outcomes of hyperacute blood pressure lowering in patients with acute ischaemic stroke within the eligible time window for reperfusion therapy.

To our knowledge, the TRUTH study is the only prospective, real-world, observational, cluster-based study to compare patient outcomes according to different blood pressure-lowering strategies before the initiation of intravenous thrombolysis.[31] This Dutch study compared the outcomes of 853 patients admitted to 27 hospitals in which clinicians used intravenous antihypertensive agents to lower blood pressure to less than the guideline systolic threshold of 185 mm Hg before administration of intravenous alteplase. The management of 199 patients from ten hospitals followed a strategy of withholding intravenous antihypertensive agents to lower blood pressure and proceeding with intravenous alteplase only if blood pressure fell spontaneously to under the guideline thresholds. Although patients who received the active blood pressure-lowering strategy had a higher rate of intravenous thrombolysis than those who did not (94% *vs* 52%; p<0·001), the study showed a trend towards harm in the group who received active blood pressure lowering versus those who did not (OR 1·27 [95% CI 0·96–1·68] after adjustment for age, sex, stroke severity, endovascular thrombectomy, baseline imbalances as fixed-effect variables, and stroke centre as a random-effect variable). However, some limitations of this observational study include the potential for selection bias and residual confounding from unaccounted interhospital variations in systems of care.

See Online for appendix

To our knowledge, there are no randomised trials of controlling blood pressure specifically in patients before they undergo endovascular thrombectomy. However, in an individual patient data meta-analysis of seven randomised trials involving 1753 patients with acute ischaemic stroke of endovascular thrombectomy for large-vessel occlusion of the anterior circulation, there was no interaction between the blood pressure on presentation and the effect of the treatment on functional outcome (p=0·96).[33] The authors subsequently suggested that endovascular thrombectomy should not be delayed in order to control blood pressure beyond the guideline threshold.[33] However, because most patients were actively treated with antihypertensive agents to lower blood pressure to less than the guideline threshold and only 92 patients were included in the analysis with a systolic blood pressure higher than 185 mm Hg before endovascular thrombectomy,[33] there are concerns over the generalisability of these findings to the outcomes of patients without treatment for very high blood pressure on presentation.

In summary, we suggest that the lowering of systolic blood pressure to 140 mm Hg or less should be avoided before initiating reperfusion therapy in acute ischaemic stroke. Moreover, any action to lower systolic blood pressure to 185 mm Hg or less, as recommended in guidelines, should be undertaken with caution, including with intravenous antihypertensive treatment to produce a rapid drop in blood pressure. There is now a pressing need for a trial of control of blood pressure to establish if it is necessary to actively lower blood pressure to less than the guideline threshold in patients before reperfusion therapy.

The management of blood pressure after reperfusion therapy

The likelihood of there being any benefits of lowering systolic blood pressure to less than 180 mm Hg in the

first 24 h after intravenous thrombolysis for acute ischaemic stroke[15–17] is probably modest at best. As the first randomised controlled trial to test the effectiveness of a lower treatment target, the ENCHANTED trial showed that more intensive lowering of systolic blood pressure to less than 140 mm Hg within 1 h post-thrombolysis reduced the odds of serious intracerebral haemorrhage, but had no effect on functional recovery.[34] A post-hoc analysis showed that, in the subgroup of patients with more severe stroke, more intensive blood pressure lowering resulted in greater mortality.[35] This finding supports the hypothesis that active lowering of blood pressure can compromise cerebral blood flow when there is a high likelihood of impaired cerebral autoregulation of blood flow. Conversely, patients with milder neurological impairment (scores ≤7 on the National Institute of Health Stroke Scale) seem able to adequately adjust to potential treatment-induced hypoperfusion.[35] An observational study showed that patients with a systolic blood pressure less than 148 mm Hg over the first 24 h after thrombolysis that was maintained at 127–138 mm Hg over the next 5 days were associated with a favourable functional outcome.[36] Although this association between a reduction in blood pressure (whether controlled or spontaneous) and improved outcomes requires cautious interpretation, these findings suggest that the management of post-thrombolysis blood pressure could necessitate dynamic temporal adjustments.

Endovascular treatments for acute ischaemic stroke continue to increase with an expansion of indications and improved technology, yet up to 50% of patients still do not have a good recovery, despite a good radiological measure of vessel reperfusion, due to complications of intracerebral haemorrhage, cerebral oedema, and reperfusion injury.[5] Many clinicians believe that patients who have successful endovascular thrombectomies might derive additional benefits from intensive control of blood pressure.[37,38]

To date, six completed randomised controlled trials have been undertaken to establish the effectiveness and harms of more intensive blood pressure lowering (systolic blood pressure <140 mm Hg) in patients with successful endovascular thrombectomy for acute ischaemic stroke from large-vessel occlusion.[26–30,39] The BP-TARGET trial[39] randomly assigned 324 patients to an intensive group with a target systolic blood pressure of 100–129 mm Hg (achieved mean 128 mm Hg) and a standard-care group with a target systolic blood pressure of 130–185 mm Hg (achieved mean 138 mm Hg). There was no significant effect of more intensive blood pressure lowering on the primary outcome of intraparenchymal haemorrhage at 24–36 h as a measure of reperfusion injury, nor was there any significant between-group differences across the various secondary outcomes (related to neurological impairment, functional recovery, serious cerebral haemorrhage, and death).[39] However, three subsequent trials conducted in countries in Asia (ENCHANTED2/MT,[26] OPTIMAL-BP,[27] and IDENTIFY[30]) were all stopped prematurely on the recommendations of their data safety monitoring boards because of the emergence of harmful effects of more intensive blood pressure-lowering treatments. The ENCHANTED2/MT study[26] randomly assigned 821 patients in China to a more intensive group (target systolic blood pressure <120 mm Hg; mean achieved 121 mm Hg) and a less intensive group (target systolic blood pressure 140–180 mm Hg; mean achieved 139 mm Hg). The between-group adjusted mean difference in systolic blood pressure was –18 mm Hg (95% CI –19 to –17). The result was a higher odds of worse functional outcome according to mRS score in the more intensive group compared with the less intensive group (common OR 1·37 [95% CI 1·07–1·76]). The adverse effect of more intensive blood pressure lowering emerged at 24 h post-randomisation, with a higher rate of death or neurological deterioration in the more intensive group (1·53 [1·18–1·97]) at 7 days compared with the less intensive group. There was no significant difference in the rate of intracerebral haemorrhage between the two groups.[26]

Similar adverse effects of more intensive blood pressure lowering were seen in the OPTIMAL-BP trial in South Korea,[27] in which more intensive treatment (target systolic blood pressure <140 mm Hg; achieved mean 129 mm Hg) were compared with less intensive treatment (target systolic blood pressure 140–180 mm Hg; achieved mean 138 mm Hg). Among the 305 patients recruited, the proportion with a good functional outcome at 90 days was significantly less in the intensive group (39·4%) than in the less intensive group (54·4%; adjusted OR 0·56 [95% CI 0·33–0·96]; p=0·03). There was no significant difference in symptomatic intracranial haemorrhage between the two groups. Finally, the IDENTIFY trial,[30] also conducted in China, recruited 383 patients after endovascular thrombectomy within 6 h of the onset of symptoms to be managed according to systolic blood pressure targets of less than 130 mm Hg in an intensive group versus less than 180 mm Hg in a usual-care control group. The result was a higher proportion of unfavourable functional outcome (mRS scores 3–6) in the intensive group (71%) compared with the control group (67·5%; relative risk 1·05 [95% CI 0·92–1·20]; p=0·45). There was no significant difference between the groups at 90 days in symptomatic intracerebral haemorrhage, malignant brain oedema, or all-cause death.[30]

There have been two other smaller pilot feasibility trials: the BEST-II trial in the USA[28] and the DETECT trial in Canada.[29] The BEST-II trial[28] randomly assigned 120 patients with successful endovascular thrombectomy to three groups with different systolic blood pressure treatment targets of less than 140 mm Hg, 160 mm Hg, and 180 mm Hg. The mean achieved systolic blood pressure across the three groups over 24 h were

122 mm Hg (target <140 mm Hg), 130 mm Hg (target 160 mm Hg), and 129 mm Hg (target 180 mm Hg). The primary aim of BEST-II was to test the feasibility of a future trial according to different degrees of treatment effects from intensive blood pressure lowering. The secondary aim was to test the futility of identifying the effect of the two lower blood pressure-lowering treatment targets compared with the higher treatment target on infarct volume at 36 h and scores on the utility-weighted mRS at 90 days. The results did not support a futility conclusion but they did suggest that a future trial would require several thousand participants for a treatment effect to be detected from a modest effect of blood pressure lowering on functional recovery after successful thrombectomy.[28] The other trial, DETECT,[29] showed no differences between intensive blood pressure lowering (systolic target <140 mm Hg) compared with standard management (systolic target <180 mm Hg) across any exploratory endpoint.

Key characteristics and the outcomes of these trials are shown in the appendix (pp 4–6). Figure 1 presents the pooled analysis of the 1941 included patients to show that intensive blood pressure lowering had less likelihood of a favourable outcome in both a shift (common OR 0·74 [95% CI 0·62–0·87]; I^2=0·0%; p for Cochran Q=0·74; z=–3·57 [p=0·0004]) and binary (mRS scores 0 to 2; OR 0·69, 95% CI 0·57–0·84; I^2=9·0%; Cochran's Q test p=0·36) analysis of functional outcome, and with low levels of heterogeneity and publication bias (figure 2) across the trials. Moreover, there was no significant heterogeneity across different levels of between-group separation in systolic blood pressure between treatment groups across the trials (p=0·47; τ^2=0·012; figure 3), providing no evidence that the treatment effect varied with the amount of blood pressure reduction.

These results are consistent in indicating that intensive (systolic blood pressure <140 mm Hg) blood pressure-lowering treatment should be avoided in patients after successful reperfusion for acute ischaemic stroke. It is likely that the treatment induces ischaemia reperfusion injury,[40,41] despite there being no clear changes in the rates of intracerebral haemorrhage or size of cerebral infarction on follow-up brain imaging.[26,39] The underlying pathophysiological mechanisms behind these adverse effects are likely to be complex regarding the approaches taken for treatment, degree of reperfusion, size of the ischaemic penumbra, amount of collateral cerebral blood flow, and health of the microcirculation, as well as the limitations of brain imaging in quantifying cerebral injury. Even so, early control of elevated blood pressure is likely to reduce blood pressure variability, which has prognostic importance.[42] Blood pressure variability is characterised by haemodynamic instability through two crucial dimensions: the magnitude–velocity product of the speed of reduction in blood pressure and the integrated burden of prereduction hypertension exposure.[42] An individual patient data meta-analysis of 2640 patients from five observational studies[43] and a secondary analysis of the BEST study[44] both indicated that blood pressure variability after endovascular thrombectomy is an independent predictor of poor functional outcome.

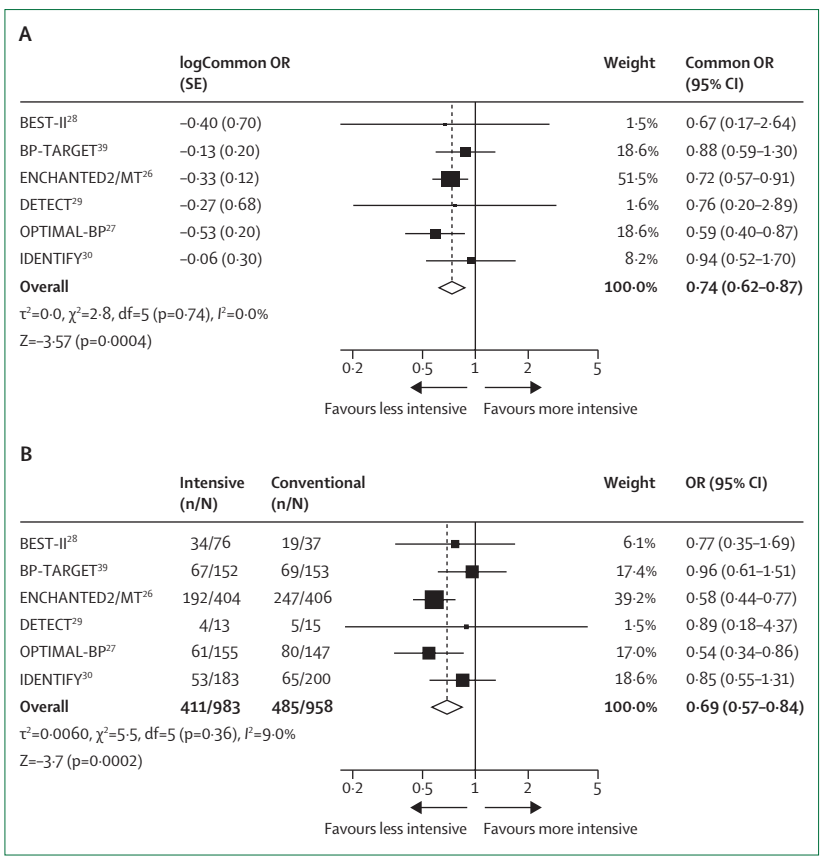

Figure 1: **Forest plots of meta-analysis of outcomes in trials of more intensive versus less intensive blood pressure lowering**

Analysis in 1941 patients who had endovascular thrombectomy for acute ischaemic stroke from large-vessel occlusion. The models used a random effect. (A) Outcome is a shift in scores on the mRS. Pooled analysis on functional improvement is defined as the score decreases on the mRS. (B) Outcome is good functional recovery defined by scores 0–2 on the mRS. Pooled analysis on good functional outcome is defined as a mRS of ≤2. mRS=modified Rankin scale. OR=odds ratio.

Conclusions and future directions

We must develop reliable randomised evidence on the control of blood pressure in patients who are eligible for intravenous thrombolysis for acute ischaemic stroke. The essential questions are: should blood pressure be altered at all and, if so, to what degree?

Immediate research gaps and priorities

According to the results of the three prehospital trials of treatment to lower elevated blood pressure in patients with suspected acute stroke and one prospective observational system-of-care study, the evidence is robust enough to indicate there is harm from intensive lowering control of blood pressure (to a systolic blood pressure <140 mm Hg). This evidence has also created uncertainty

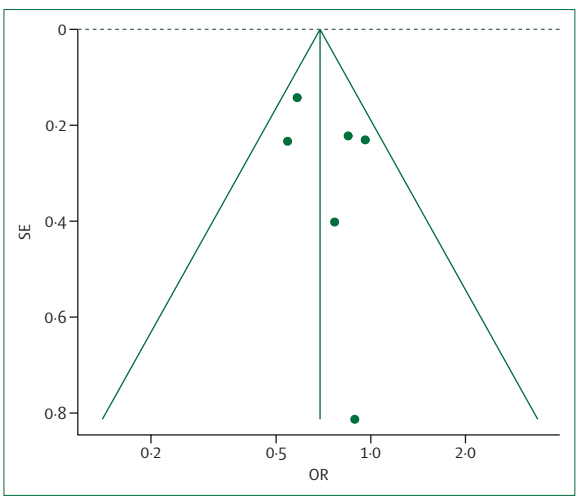

Figure 2: **Funnel plot of six published trials of blood pressure-lowering treatment in patients after successful endovascular treatment for acute ischaemic stroke**
The funnel plot visually assesses the presence of publication bias. The central line represents the pooled estimated effect (OR) and; the diagonal lines represent the expected distribution of study results in the absence of bias. OR=odds ratio.

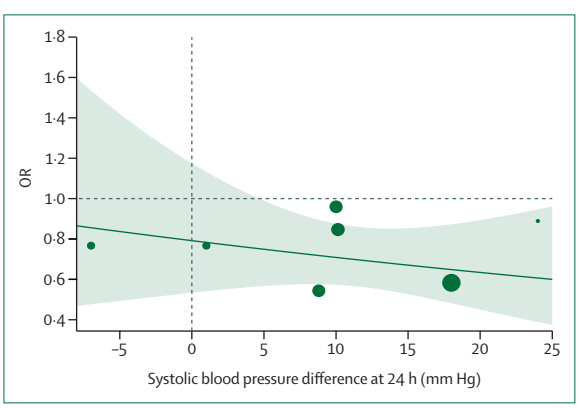

Figure 3: **Meta-regression of degrees of separation in systolic blood pressure between treatment groups in the six blood pressure-lowering trials**
The solid line represents the fitted regression line, showing how the OR is predicted to change as the difference in systolic blood pressure changes. The shaded area represents the 95% prediction interval. The horizontal dashed line represents an OR of 1·0. The circles represent the individual studies, with the size representing the statistical weight of the study in the analysis. OR=odds ratio.

as to whether a milder intensity of blood pressure control is also harmful (any pressure down to 140 mm Hg, with the conventional recommended systolic pressure being 185 mm Hg) before initiating intravenous thrombolysis in patients with acute ischaemic stroke.[45] Moreover, there appears to be no strong rationale to delay the initiation of intravenous thrombolysis to control elevated blood pressure.[33] It must be established whether such treatments should be done concurrently with or directly after thrombolysis. Despite the challenges, an international, investigator-initiated, collaborative, randomised controlled trial is warranted to definitively establish the optimal management of blood pressure before intravenous thrombolysis, potentially with adaptive elements to the design to allow comparisons between different management strategies, agents, and patients.

In patients who have received reperfusion therapy for acute ischaemic stroke, the evidence is weak for any benefit of intensive blood pressure lowering treatment. In fact, intensive control of blood pressure to a systolic target of less than 140 mm Hg appears harmful in patients who receive reperfusion therapy with endovascular thrombectomy for acute ischaemic stroke due to large-vessel occlusion.[26,27,30] However, there are two ongoing trials that aim to further test the effectiveness of intensive blood pressure lowering in patients after successful thrombectomy. The HOPE trial[46] is being undertaken in Spain and more than half of the planned 800 patients have been recruited to date, with no safety concerns from the data and safety monitoring board. The ENCHANTED3/MT trial (NCT06352619) is being done as a domain of the ACT-GLOBAL trial (NCT06320431 and NCT06352632), in which up to 2000 patients are planned to be recruited to test three different approaches to the control of blood pressure: minimal, moderate, or intensive systolic blood pressure reduction strategies.

The OPTEC collaboration, hosted by The George Institute for Global Health (Sydney, NSW, Australia), is a planned individual patient data meta-analysis of completed trials in endovascular therapy, which will allow for investigation of the interaction of blood pressure-lowering treatment and patient-level characteristics.[47] Because a dynamic change in blood pressure is usually reported to occur early after successful reperfusion and even normalise within 48 h in more than half of patients, a new hypothesis that requires testing is that a favourable outcome will occur by simply allowing blood pressure to decline spontaneously. Further details of these ongoing trials and individual patient data meta-analysis are provided in the appendix (pp 7–8).

Long-term research gap and priorities

An individualised approach to the management of blood pressure might be required in the future to account for a patient's baseline blood pressure, pathological type of cerebral ischaemia, clinical severity, cerebrovascular reserve, degree of vascular reperfusion, and other clinical and imaging features. At present, however, to our knowledge, no analysis in any trial has provided clear evidence of heterogeneity in the effects of blood pressure lowering across demographic, clinical, medical history, or treatment variables. There is interest in non-invasive methods of monitoring, such as through transcranial doppler or near-infrared spectroscopy,[48] which might allow for real-time adjustments of blood pressure at the bedside to guide the development of individualised blood pressure management around

Search strategy and selection criteria

We searched the Cochrane Library, MEDLINE, and Embase for articles published in English between Nov 1, 2022, and May 31, 2025. We used search terms of "ischaemic stroke", "reperfusion", "intravenous thrombolysis/thrombolysis", "endovascular thrombectomy/endovascular treatment/mechanical thrombectomy/thrombectomy", and "clinical trials" or "meta-analysis". We searched the references identified by all the search strategies and evaluated the articles we considered relevant. We primarily selected the results of clinical trials that were published within the past 24 months but did not exclude articles that were widely cited and highly regarded. We included review articles to provide readers with more details and references than those included in this Rapid Review (details in the appendix pp 2–8).

safety parameters. Artificial intelligence based on deep learning could be an effective solution to find the threshold values of these individualised blood pressure parameters.

Future research is warranted according to the presumed mechanism for the effects of blood pressure on cerebral blood flow and function. A secondary analysis of ENCHANTED2/MT showed that the harmful effects of intensive blood pressure lowering treatment were strongly modified by the degree of collateral blood flow, but were not modified in relation to other brain imaging parameters, such as the size of cerebral infarction or degree of reperfusion after endovascular therapy.[49] This finding indicates that imaging biomarkers could provide a basis for individualising the management of blood pressure. Finally, consideration should be given to various vascular regulation factors, including the ability of the brain to self-regulate regarding causal mechanisms of cerebral ischaemia, degrees of collateral circulation, and success of reperfusion.

Contributors

LS and CSA conceptualised the report. LS and YM searched the literature and drafted the manuscript. All authors reviewed and edited the manuscript.

Declaration of interests

LS has received grants from the UK Medical Research Council and Unionstrong Technology; is a member of the data and safety monitoring board (DSMB) for the PROTECT-MT, MILD-MT, PATH-ICAS, and TNK-MeVO trials; and is the co-principal investigator (PI) for INTERACT4 trial. NDK is a co-PI of the TRUTH study and has received a grant from Fonds NUTS-OHRA. PN has received grants from the Netherlands Heart Foundation/CardioVascular Onderzoek Nederland and Fonds NUTS-OHRA; is the co-PI of the MR-ASAP and TRUTH studies; and is a member of the DSMB for the TORCH trial. CSA has received grants from the National Health and Medical Research Council of Australia, the UK Medical Research Council and Medical Research Foundation, and AstraZeneca, paid to his institution; reports receiving advisory committee fees from AstraZeneca and Auzone; is Vice-President of the World Stroke Organisation; is Editor-in-Chief of *Cerebrovascular Diseases*; is Associate Editor of the *International Journal of Stroke*; is co-PI for the INTERACT4, ENCHANTED, ENCHANTED2/MT, and ENCHANTED3/MT clinical trials; and is a member of the DSMB for the PROTECT-MT, MAGIC-MT, and DIST trials. All other authors declare no competing interests.

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